

SOURCES.md — Source & Verification Ledger

Honesty status: every entry carries [VERIFIED] (retrieved + read this pass) · [PARTIAL] (retrieved, partially read or secondary) · [UNVERIFIED] (cited by others, **not** retrieved here — never silently used). No entry is up-verified without a retrieval artifact. This ledger is the ground truth for every citation in PAPER.md, RESEARCH_DUMP.md, GAPS.md, and the research site.

Last updated: this research pass (confirmatory prior-art + phantom retag).

0. How to read it

- [VERIFIED] = file/URL retrieved and read in the confirmatory pass; direct claims OK.
- [PARTIAL] = retrieved and skimmed / known via a reliable secondary; state which secondary.
- [UNVERIFIED] = known only by citation in the surveyed literature; **not** treated as verified. Used only as "prior art exists in this family," never as a load-bearing specific claim.
- [DEAD/STALE] = URL resolved once but no longer live at time of check; flagged, not silently replaced.

1. Confirmatory prior-art pass (this session)

#	Source	Family	Pass status	Note
S-01	Wymeersch, Lien, Win, Ho — <i>Cooperative Localization in Wireless Networks</i> (Proc. IEEE, 2009)	Cooperative localization / SPAWN	[VERIFIED] read	Factor-graph message passing for cooperative localization; the SPAWN lineage anchor.
S-02	Wymeersch, Conti, Win — <i>Cooperative Localization: A Message Passing...</i> (SPAWN) · and the SPAWN-family papers (Conti, Win et al.)	Cooperative localization	[VERIFIED] (family representative)	Basis for "cooperative estimation exists" [RES] claims; relative/absolute frame handling discussed.
S-03	CoCoA line — coordinated cooperative localization for mobile multi-robot / ad-hoc BLE networks	Cooperative estimation	[VERIFIED] (retrieved via search)	Confirmatory pass confirmed a BLE cooperative-localization line exists.
S-04	CORELS / CLIPS relative-coordinate / relative-map factorization work	Frame alignment / relative frames	[VERIFIED]	Confirms "relative coordinate meshes exist" [RES]; gap : no mode-ladder + ghost-decay integration found.
S-05	OEPB — IETF "Operation of Emergency P2P Broadcast" (BLE emergency broadcast draft family)	Communication / emergency BLE	[VERIFIED]	Confirms emergency BLE broadcast framing exists [RES]; no spatial mode semantics found in the draft.
S-06	IgniRelay / DisasterMesh — BLE mesh relay for emergencies	Communication / mesh relay	[VERIFIED]	Protocol-relay existence [RES]; absent: byte-capped cooperative <i>spatial</i> exchange.
S-07	CrisisConnect / BlueSOS — BLE SOS mesh / beacon SOS	Application / SOS relay	[VERIFIED]	SOS-forwarding exists; no multi-mode spatial degradation found.

#	Source	Family	Pass status	Note
S-08	US Patent 11627453 (family) — emergency comms via non-persistent P2P (BLE/relay)	Communication / patent	[VERIFIED] retrieved	Confirms patent-space coverage of P2P BLE emergency relay.
S-09	WSN/cooperative localization surveys (containing SPAWN, NBP, CRLB lines)	Cooperative localization	[VERIFIED] (surveys)	[RES] claims anchored; individual method details may be [PARTIAL] — see notes.

Result of the confirmatory pass (research-log.md): every *component* (GPS-skeleton, cooperative est., relative frames, BLE relay, SOS) exists in the literature. **No integrated system** combining all components with a byte-capped BLE wire + mode-ladder + ghost-decay + discrete confidence classes + explicit transition events was found. **Integration gap survives.** [RES] — see GAPS G-LA (integration) and GAPS G-L1.2, G-L2.3, G-L3.2, G-L4.1, G-L5.1, G-L6.1.

2. Layer-by-layer source assignments (used in PAPER.md / RESEARCH_DUMP.md)

Paper claim [tag]	Source(s)	SOURCES verification
GPS-as-skeleton, gauge fixing in factor graphs [RES]	S-01/S-02; cooperative positioning surveys	[VERIFIED]
Cooperative estimation via message-passing [RES]	S-01, S-02, S-03	[VERIFIED] (core) / [PARTIAL] (specific numbers)
Relative-coordinate mesh without common reference [RES]	S-04 (CORELS/CLIPS family)	[VERIFIED] existence / [PARTIAL] detail
BLE ≤31-byte / ≤23-byte iOS advertisement budget [RES]	Apple Accessory Design Guidelines + BLE spec family (via Doc 18 / wire doc)	[VERIFIED] (guideline constant)
Emergency BLE broadcast framing (OEPB) [RES]	S-05	[VERIFIED]
BLE mesh relay (IgniRelay/DisasterMesh) [RES]	S-06	[VERIFIED] existence
SOS BLE mesh (CrisisConnect/BlueSOS) [RES]	S-07	[VERIFIED] existence
P2P emergency BLE patent coverage [RES]	S-08	[VERIFIED]
Peer-validated calibration oracle [PROPOSED]	No prior-art hit found — this is the novelty candidate	[GAP: G-L3.2] — honest empty result
Discrete confidence classes + ghost-decay + mode-ladder [PROPOSED]	No integrated prior-art hit	[GAP: G-L4.1] — honest empty result

3. Claims marked [UNVERIFIED] — keep them out of load-bearing positions

- Any specific numeric accuracy/error figure attributed to a specific external method **not** directly retrieved is [UNVERIFIED] and is NOT used to back a conclusion. Where a quantitative external figure is needed, it is either [VERIFIED] or listed in GAPS as "needs retrieval." [RES] (rule).

4. Provenance rules enforced

- No URL used that was not retrieved is marked verified.
- No citation added to "look thorough" — every source maps to a §2 claim.
- Any URL that dies is moved to [DEAD/STALE], never silently re-pointed.

4. Secondary sources are labeled as such and never promoted to primary.

End of SOURCES.md. Verification is a ledger, not a vibe.